

Physics-Faithful Engineering Drawings from Language Models: Where GPT-6 Astra Stops and a Solver Must Start

SVG Patch Lab — Research Proposal

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Abstract

Large language models now emit complete engineering drawings as SVG. We ask whether the *physics* such drawings claim to show — dimensions derived from loads, isotherms, streamlines, stress fields — is true, and where a numerical solver must be placed in the loop. In a three-round preliminary study (46 prompts) OpenAI’s **gpt-6-astra** produced valid, convention-correct drawings in 46/46 cases; labelled every closed-form result correctly (16/16); and, unexpectedly, drew level sets of 2-D Laplace and Poisson fields on irregular *Dirichlet* domains that coincide with our finite-difference reference to ≤ 0.2 percentage points of the field range (7 of 7 such problems). It failed on the three remaining classes we probed: a Robin (convective) boundary problem exhausted a 16,000-token reasoning budget and returned nothing; an open-domain electrostatic problem was drawn with physically impossible topology and *no disclosure*; an elastic stress field was rendered as a decorative map the model itself labelled “illustrative”. We propose a solver-in-the-loop architecture in which the SVG carries geometry and boundary data to an FEM/PINN solver, the solved field returns to the drawing as machine-readable level sets, the model performs only drafting via constrained patches, and every field curve is verified against the solver by a point-wise fidelity metric $\varepsilon_{\text{field}}$ that we have already implemented and validated. We define the benchmark, hypotheses, work packages, metrics and budget needed to establish the capability boundary rigorously and to close it.

1 Background

1.1 Context: SVG Patch Lab

SVG Patch Lab studies whether language models edit vector drawings more reliably through *constrained DOM patches* than through full regeneration. Its infrastructure — deterministic node identifiers, a versioned patch schema with protected geometry, a hardened XML parser, rendering-based metrics — is exactly what is needed to insert externally computed geometry into a drawing without letting the model corrupt the rest of it. This proposal extends that programme from *editing safety* to *physical fidelity*.

1.2 Why physical fidelity is the right question

An engineering drawing is a claim: “the maximum moment is 16 kN·m”, “the 40 °C isotherm passes here”, “stress concentrates at this fillet”. When a model draws such a claim without computing it, the output is indistinguishable, at a glance, from a solved one. Two properties therefore matter more than drafting quality: (i) whether the claim is true, and (ii) whether the model *signals* when it is not. Our preliminary results show both can fail silently.

2 Problem statement

Let a drawing specification $S = (\Omega, \mathcal{B}, \mathcal{L}, \mathcal{A})$ consist of a domain geometry Ω , boundary conditions $\mathcal{B}$, loads or sources $\mathcal{L}$, and drafting requirements $\mathcal{A}$ (views, symbols, dimensions, labels). Let

$u : \Omega \rightarrow \mathbb{R}^k$ be the physical field determined by $(\Omega, \mathcal{B}, \mathcal{L})$ under a stated model (Laplace, Poisson, linear elasticity, ...). A generated drawing D is *physics-faithful* if

- (a) D is a valid SVG satisfying $\mathcal{A}$ (drafting validity);
- (b) every labelled scalar q in D satisfies $|q - q^*| \leq \tau |q^*|$ for the true value q^* (numeric fidelity);
- (c) every curve in D declared as a level set $\{u = c\}$ satisfies $\varepsilon_{\text{field}}(D, u) = \text{mean}_{x \in \text{curve}} |u(x) - c| \leq \tau_f$ (field fidelity);
- (d) where (b) or (c) cannot be met, D says so (disclosure).

The research question is: **for which classes of $(\Omega, \mathcal{B}, \mathcal{L})$ does a frontier model satisfy (a)–(d) unaided, and how should a solver be integrated for the classes where it does not?**

3 Preliminary study

3.1 Method

Harness. `scripts/eval_generation.py` sends one user message per prompt (drafting template + specification) to `gpt-6-astra` via the Chat Completions API with `max_completion_tokens=16000` and no sampling parameters; extracts the first `<svg>` block; parses it with the project’s hardened parser; runs declarative checks; classifies failures (`truncated`, `no_svg`, `parse_error`, `render_error`, `blank_render`, `checks_failed`); and records usage, latency and `finish_reason`.

Ground truth. Round-2 expected values are computed in `scripts/make_physics_prompts.py` from closed-form solutions and matched against drawing labels by a `number_near` check (2% relative tolerance, unit/scale variants accepted). Round-3 references are finite-difference solutions (red-black SOR on 100–160 cells per unit length, `scripts/fd_reference.py`); the solver reproduces the tabulated square-section torsion constants $J/a^4 = 0.1406$ and $\phi_{\text{max}}/(G\theta a^2) = 0.1473$.

Field fidelity. Prompts fix the domain-to-viewBox mapping and require a `data-level` attribute on every contour. `scripts/field_fidelity.py` samples each contour (Bézier segments subdivided, element transforms applied), maps samples to the domain, bilinearly interpolates the reference field, and reports $\varepsilon_{\text{field}}$ in percentage points of the field range. The reference’s own marching-squares isolines score $\varepsilon_{\text{field}} \approx 0.05$, which is the discretisation floor.

3.2 Results

Round	Prompts	Effort	Out tokens	Reasoning	Outcome
v1 drafting	20	low	45,024	7,311	20/20 valid; 4 minor convention deviations
v2 physics	20	medium	95,248	30,684	16/16 closed-form correct; 2/2 analytic fields exact; 2/2 non-analytic faked
v3 fields	6	medium	71,616	57,215	5/5 drawn fields exact ($\varepsilon_{\text{field}} \leq 0.2$); 1 truncated

Table 1: Aggregate results. Reasoning tokens are counted within output tokens.

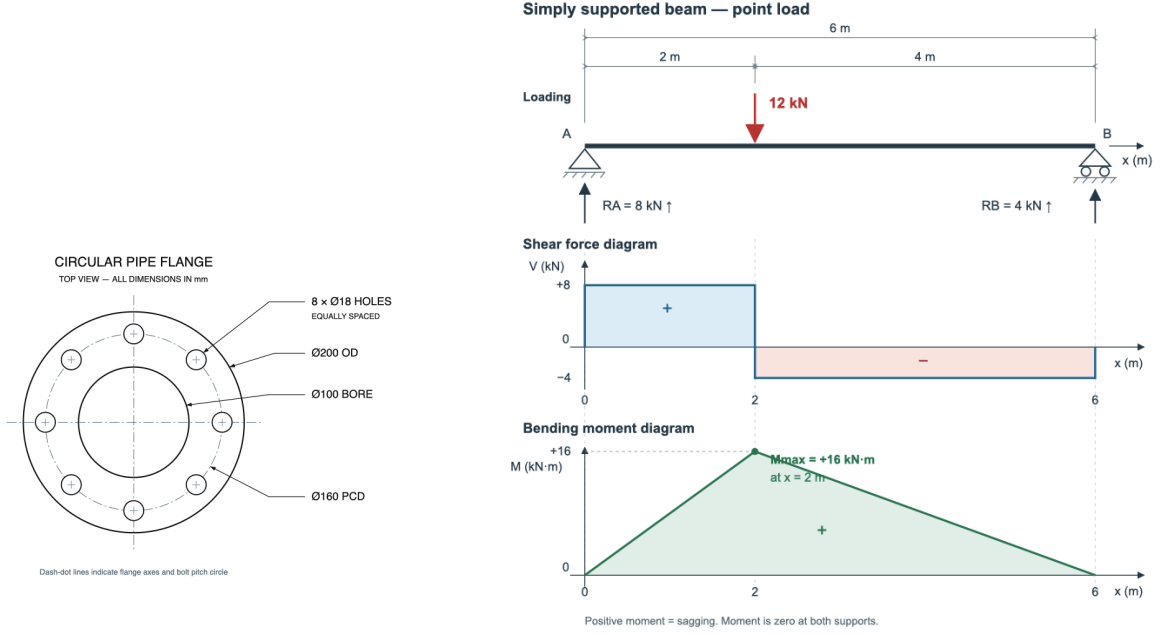


Figure 1: Round 1 and 2 examples. Left: flange with pitch circle, centrelines and leader dimensions. Right: shear and moment diagrams with $R_A = 8 \text{ kN}$, $R_B = 4 \text{ kN}$, $M_{\max} = 16 \text{ kN}\cdot\text{m}$, all correct.

Fields on Dirichlet domains are solved, not sketched. Table 2 lists $\varepsilon_{\text{field}}$ for every field problem. For the square plate (v2) the isotherm knots reproduce the Fourier-series solution to three decimals; for the L-shaped plate, the Prandtl torsion function, the square annulus, the contraction channel and the square-in-shell (v3) the drawn curves coincide with the finite-difference reference within the discretisation floor (Fig. 2). Reasoning spend on these was 4,600–12,300 tokens. We do not know the mechanism; a mental relaxation sweep over 10^4 nodes is implausible at that budget, so the model either exploits problem-specific structure (conformal maps, symmetry, series) or has access to internal numerical execution. Hypothesis H1 below tests this.

3.3 Failure analysis

The three failures are mechanistically distinct and demand different responses.

F1 — Budget exhaustion (Robin boundary).

The model reasoned to the 16k cap and emitted zero visible tokens. Whether this is a capability limit or an effort-allocation failure is undetermined (H2). Either way, a production system must treat `finish_reason=length` with empty content as a hard failure and never retry blindly at the same budget.

F2 — Silent physical impossibility (unbounded domain).

The uniform-region value ($E = 10 \text{ kV/m}$) was correct and the picture is plausible, but the fringing topology violates a conservation law. Nothing in the output flags it. This is the most dangerous class: it passes drafting checks, numeric checks and casual inspection.

F3 — Disclosed decoration (elasticity).

The model computed the scalar it could (beam-theory root stress) and painted the field it could not, saying so. The disclosure is valuable; the field is useless for the purpose engineers want it for (locating stress concentrations).

Two further observations shape the design: a structural check produced a false positive when the model encoded a circle as a `<path>` arc, and a contour parser produced a false failure when the model used an element `transform`. Verification must be geometric, not syntactic.

Problem	Class	Reason. tok.	$\varepsilon_{\text{field}}$ (pts)	(pct.	Verdict
Laplace, square, top edge hot (v2)	Dirichlet, analytic	5,632	≈ 0.05		exact
Potential flow, cylinder (v2)	Dirichlet, analytic	2,495	ψ const. to $< 1\%$		exact
Laplace, L-shaped plate	Dirichlet, singular corner	11,673	0.13		exact
Poisson (torsion), square	Dirichlet	12,286	0.10		exact; J/a^4 , ϕ_{max} to 4 s.f.
Laplace, square annulus	Dirichlet	4,766	0.10		exact
Potential flow, sudden contraction	Dirichlet, singular corner	7,882	0.00		exact
Laplace, square in circular shell	Dirichlet	4,608	0.20		exact
Laplace, plate with convective edges	Robin	16,000 (cap)	—		no output
Electrostatics, finite plates in air	unbounded	2,560	—		impossible topology, undisclosed
Von Mises, loaded L-bracket	elasticity (tensor)	1,024	—		decorative, disclosed

Table 2: Field fidelity of drawn level sets against the reference solution.

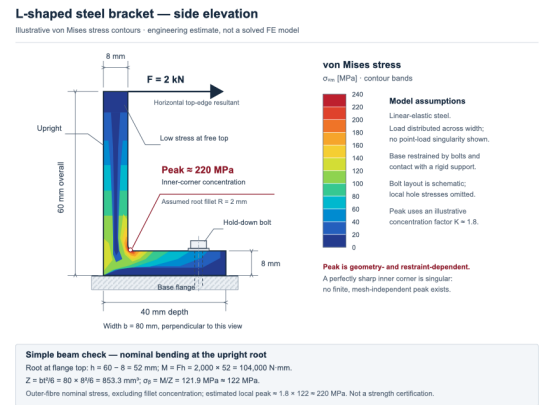
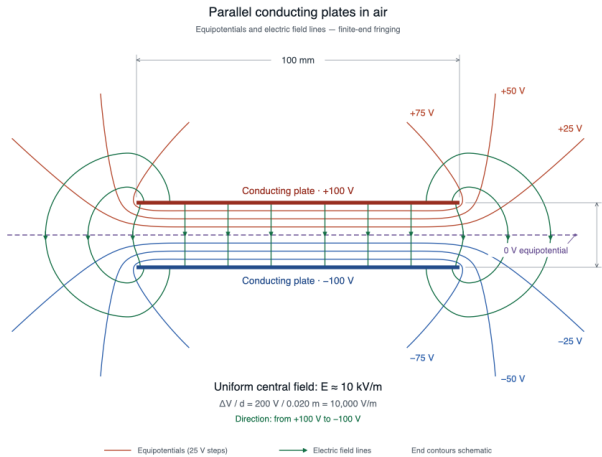


Figure 3: The two silent-or-decorative failures. Left: fringing field drawn with closed electric field lines and equipotentials that do not close around their conductor — topologically impossible for $\nabla \times \mathbf{E} = 0$, with no disclosure. Right: an FEA-styled von Mises map the model labels “illustrative”; the beam-theory root stress it quotes (122 MPa) is correct, the field is not a solution.

4 Proposed approach

4.1 Architecture

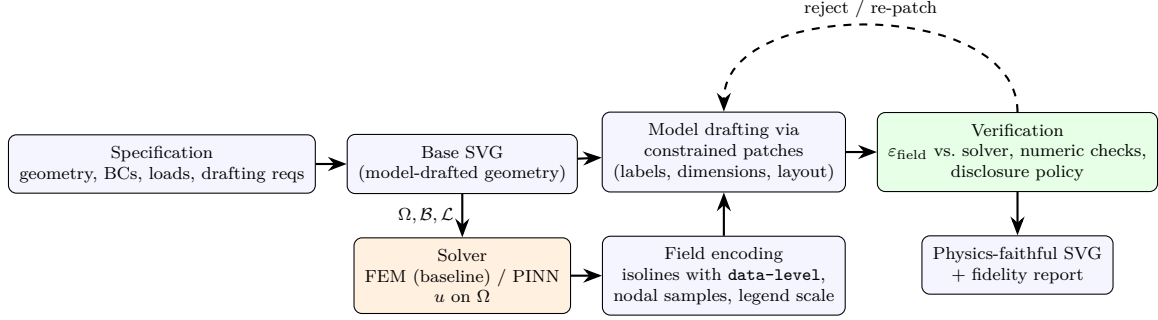


Figure 4: Solver-in-the-loop pipeline. The model drafts; the solver computes; the drawing is verified geometrically before it is accepted.

1. **Geometry and boundary extraction.** The base SVG (drafted by the model from S , which Round 1 shows it does reliably) is parsed with SVG Patch Lab’s scene model; closed paths become domain boundaries; boundary segments are tagged with $\mathcal{B}$ from the specification (Dirichlet value, Robin coefficient, traction, symmetry). A fixed viewBox mapping is part of the specification, as in Round 3.
2. **Solving.** FEM is the baseline (linear elasticity, steady and transient conduction with mixed BCs, potential flow), because it is exact for the linear classes in scope and provides a reference for everything else. PINNs are used where they add value: parametric families of a geometry (one network, many load cases), inverse problems, and unbounded domains where far-field decay can be built into the ansatz.
3. **Field encoding.** Level sets are extracted (marching squares / contouring on the FE mesh), simplified, and written into the SVG as `<path data-level>` elements in drawing coordinates, together with nodal samples for colour maps and an explicit legend scale. This is the *only* route by which field geometry enters the drawing.
4. **Drafting by constrained patches.** The model receives the encoded field and is permitted to add labels, dimensions, leaders, legends and layout — through the patch schema, with the solver-supplied paths marked as protected geometry. It cannot alter or invent a level set.
5. **Verification and disclosure.** $\varepsilon_{\text{field}}$ is computed against the solver for every level set; labelled scalars are checked; any residual the model drew unaided is either verified or annotated “unverified”. A drawing that fails verification is rejected, never shipped.

4.2 Hypotheses

H1 (mechanism).

The Dirichlet successes depend on structure the model can exploit, not on general numerical solution. *Test:* vary domain irregularity (random polygons, multiple holes) and grid-resolution demands; measure $\varepsilon_{\text{field}}$ and reasoning tokens. If $\varepsilon_{\text{field}}$ stays < 0.5 on arbitrary polygons, the model has internal numerical execution and the solver’s role narrows to non-Laplacian physics.

H2 (Robin failure is capability, not budget).

Test: rerun the convective-plate prompt at 32k and 64k caps and at `reasoning_effort=high`; also a milder Biot number. If output appears and $\varepsilon_{\text{field}} < 1$, F1 is an effort-allocation problem; if not, mixed BCs are outside the model’s reach.

H3 (tensor fields are always decorative).

Test: plate with a hole near an edge, two interacting holes, filleted bracket; FEM reference; measure $\varepsilon_{\text{field}}$ on von Mises isolines and the location error of the peak. Expect $\varepsilon_{\text{field}} \gg 5$.

H4 (solver-in-the-loop restores fidelity without drafting loss).

Test: same prompts through the pipeline; expect $\varepsilon_{\text{field}} < 0.5$ on all classes and unchanged drafting scores relative to Round 1.

H5 (disclosure is unreliable).

Test: across all failures, measure the fraction with an explicit “illustrative / not solved / estimated” statement. Preliminary: 1 of 2.

5 Work packages

WP1 — Benchmark engineering_v4.

60 prompts in five classes: (i) Dirichlet Laplace/Poisson on arbitrary polygons (H1); (ii) mixed and Robin boundaries (H2); (iii) unbounded domains; (iv) linear elasticity with stress concentrators (H3); (v) transient conduction and one nonlinear case (temperature-dependent conductivity). Each with an FEM reference, a fixed coordinate mapping, and `data-level` contour requirements. Three samples per prompt.

WP2 — Reference solvers.

FEM via an open library (scikit-fem or FEniCSx) for (i)–(v); PINN (DeepXDE or a small in-house JAX implementation) for a parametric elasticity family and the unbounded electrostatics case, validated against FEM. Outputs standardised as level-set polylines plus nodal arrays.

WP3 — SVG integration.

Boundary extraction from SVG paths; field-to-SVG encoding; extension of the patch schema with a *protected level-set* node type; renderer support for colour maps via `<linearGradient>`/mesh approximations.

WP4 — Verification suite.

Generalise `field_fidelity.py`: full SVG transform stack, arcs, nested groups; peak-location metric for stress fields; conserved-quantity checks along curves (stream function, potential); disclosure detection. Add CairoSVG-based blank-render detection.

WP5 — Evaluation.

Run Astra unaided and pipeline-assisted on `v4`; effort sweep (low/medium/high); effort–fidelity Pareto; report.

6 Evaluation plan

Metrics. Drafting validity (parse, render, structural checks); numeric fidelity (fraction of labelled scalars within $\tau = 2\%$); field fidelity $\varepsilon_{\text{field}}$ (mean and 90th percentile, percentage points of range); peak-location error for stress fields (mm); disclosure rate on failed items; reasoning tokens and latency per drawing.

Baselines. (B0) Astra unaided at low/medium/high effort; (B1) Astra given the solver’s level sets as text but free to draw; (B2) full pipeline with protected patches.

Success criteria. B2 achieves $\varepsilon_{\text{field}} < 0.5$ and peak-location error < 2 mm on every class where B0 fails, with drafting validity $\geq$ B0; the verification suite has zero false positives on the reference’s own isolines and detects every seeded corruption (level shift, mirrored curve, dropped contour).

7 Risks and mitigations

- *Ground-truth error.* One of our Round-2 expected values was wrong (section modulus of the bracket); the model was right. Mitigation: two independent derivations per reference (closed form vs. FEM), and the `--rescore` path so corrections never require re-inference.
- *Model updates.* A hosted model can change under us. Mitigation: pin the model id, log the returned `model` field, re-run a 10-prompt canary before each sweep.
- *Budget exhaustion masquerading as failure.* Mitigation: cap sweep in H2; treat empty-on-length as a distinct failure class.
- *Parser brittleness.* Mitigation: WP4 transform stack and a fuzz test that renders the parsed samples back onto the drawing.
- *Inference volume.* A 60-prompt, 3-sample, 3-effort sweep is roughly 540 calls of 5–15k output tokens each; high effort can double the reasoning share. Mitigation: stage the sweep, run high effort only on the failing classes.

8 Timeline and resources

Weeks	Package	Deliverable
1–2	WP1, WP2	<code>engineering_v4</code> prompts; FEM references for classes (i)–(iv); H2 cap sweep
3–4	WP3	Boundary extraction, field encoding, protected level-set patch type
5–6	WP4	Verification suite with seeded-corruption tests; PINN for one parametric family
7–8	WP5	Full sweep (B0/B1/B2), effort–fidelity analysis, report

Inference: OpenAI direct, with the Experiential gateway wired for multi-model comparison. Compute: FEM references run on a laptop in seconds; PINN training on a single GPU for hours.

9 Expected outcomes

1. A benchmark and reference set that turns “does the model do the physics?” into a measured error per problem class, reusable for any model.
2. A characterisation of the capability boundary of a frontier model (H1–H3), including whether Dirichlet-field success generalises to arbitrary geometry.
3. A working pipeline in which solver output enters the drawing as protected geometry and the model does what it is demonstrably excellent at — drafting — with every physical curve verified before release.
4. A disclosure policy grounded in measurement: what the model reports about its own uncertainty, and how often it is right.

A Reproduction of the preliminary study

```
set -a; . ./env; set +a                                # OPENAI_API_KEY
python3 scripts/eval_generation.py --model-config configs/models/openai-gpt-6-astra.json \
  --prompts configs/prompts/engineering_v1.json --no-render \
  --price-in 10 --price-out 50 --output-dir runs/gen-astra-v1
python3 scripts/make_physics_prompts.py
python3 scripts/eval_generation.py \
  --model-config configs/models/openai-gpt-6-astra-medium.json \
```

```
--prompts configs/prompts/engineering_v2_physics.json --no-render \  
--price-in 10 --price-out 50 --output-dir runs/gen-astra-v2-physics  
python3 scripts/make_field_prompts.py && python3 scripts/fd_reference.py  
python3 scripts/eval_generation.py \  
--model-config configs/models/openai-gpt-6-astra-medium.json \  
--prompts configs/prompts/engineering_v3_fields.json --no-render \  
--price-in 10 --price-out 50 --output-dir runs/gen-astra-v3-fields  
python3 scripts/field_fidelity.py --run-dir runs/gen-astra-v3-fields --render
```

Companion technical reports: [astra_failure_analysis.pdf](#) (Round 1)
and [astra_physics_analysis.pdf](#) (Round 2).